

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics

Coursework: Probabilistic Artificial Intelligence, Advanced Machine Learning, Image Analysis and Computer Vision, Planning and Decision Making for Autonomous Robots, 3D Vision, Perception and Learning for Robotics, Deep Generative Models, Robot Learning

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering

Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

Knowledge Distillation for Aerial Robots Control  Video  Report Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab

- Multi-teacher distillation for the OMAV, an overactuated tilt-rotor MAV: each of 30,000 randomized simulated vehicles is flown by its own parameter-matched expert, and the whole expert family is compressed into one LSTM policy trained closed-loop by backpropagating through the differentiable vehicle dynamics (BPTT)
- Outperforms the nominal model-based controller in velocity tracking (2,000 paired rollouts) and nearly halves its divergence rate under $\pm 30\%$ model mismatch; deployed zero-shot on the real OMAV

Goal-Driven Adaptive-Resolution 3D Mapping with RL  Report Feb–Jun 2026
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Dense 3D maps waste memory on detail the task never uses – trained a PPO agent that decides online, region by region, where the map needs fine voxels and where coarse ones suffice, under a hard memory budget
- On unseen scenes the learned map keeps 79% of point-goal success with 87% less memory than a uniformly fine map (20-episode validation); built as a sparse hierarchical rewrite of PEANUT's voxel map (Habitat, HM3D)

Selected Projects

Vision-Language-Action Policies on a SO-101 Arm  Repo 05/2026
Project of the Class Robot Learning

- Fine-tuned SmoVLA policies to follow written commands – “place the can on the person left of Obama” – generalizing to celebrity faces never seen in training, on-robot at 10 FPS; built the LeRobot data pipeline and H100 cloud training

DINOv3 for Mitral Valve Segmentation  Repo 12/2025

- Clinicians annotate only a few frames per medical video – propagated those sparse labels through whole videos via DINO-based feature propagation to generate a full pseudo-labeled training set, then trained a U-Net segmentation model on it

Multi-Agent Multi-Goal Collection Planner 12/2025
Project of the Class Planning and Decision Making for Autonomous Robots

- A fleet of robots divides up collection goals among themselves (Large Neighborhood Search), plans paths in real time, and avoids each other with no central coordinator – each robot seeing only part of the map

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo Feb–Jun 2025
Research Intern — Industrial Robotics  Repo

- Took a linear actuator from CAD design to real-time Arduino control, validated in MATLAB simulation

ETH Robotics Club Nov 2025–Present
Partnerships Team Lead

- Secured industry sponsorships for RoboHack 2026, the club's flagship robotics hackathon

Laelaps AI Jun 2026–Present
Robotics Engineer

- Developing a multi-agent RL (MARL) framework for scalable patrolling of large areas with robot teams